NPTEL ONLINE CERTIFICATION COURSES

Education for Sustainable Development

Prof. Atasi Mohanty
Rekhi Centre of Excellence for the Science of Happiness, IIT Kharagpur

Lecture 01: Introduction

LECTURE # 01

Education for Sustainable Development

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KEYWORDS

- Education
- Sustainability
- Sustainable Development
- Sustainable Development Goals
- Education for Sustainable Development

Course objectives

- Students will gain understanding of Sustainable Development and its 17 goals
- Students will be able to understand and explain SDG 4- Quality Education for all
- Students will be able to analyze how to use Education for Sustainable Development to achieve SDG 4.7 and SDG 4.4
- Students will be able to study and link sustainable education and global partnership

Expected Learning Outcomes

- By the end of the course students will be able to:
- Describe the UNESCO's 17 Sustainable Development Goals
- Identify elements of Education for Sustainable Development
- Illustrate and integrate ESD to achieve SDG 4.7
- Explain and formulate ESd to accomplish SDG 4.4
- Interpolate sustainable education and global partnership

REFERENCES

https://sdgs.un.org/goals
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https://www.youtube.com/watch?v=1zm7In9FRtE&t=4s

SUSTAINABLE DEVELOPMENT

A HISTORICAL INTRODUCTION TO UNESCO 17 SUSTAINABLE DEVELOPMENT GOALS

SUSTAINABLE DEVELOPMENT GOALS

WHAT ARE THE SUSTAINABLE DEVELOPMENT GOALS?

- **Purpose :** The Sustainable Development Goals (SDGs) also known as the Global Goals, were adopted by all United Nations Member States in 2015 as a universal call to action to end poverty, protect the planet and ensure that all people enjoy peace and prosperity by 2030.

WHAT ARE THE SUSTAINABLE DEVELOPMENT GOALS?

- **Aim** : The SDGs are a bold commitment to tackle some of the more pressing challenges facing the world today. All 17 Goals interconnect, meaning success in one affects success for others. Dealing with the threat of climate change impacts how we manage our fragile natural resources, achieving gender equality or better health helps eradicate poverty, and fostering peace and inclusive societies will reduce inequalities and help economies prosper.

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ORIGIN OF SUSTAINABLE DEVELOPMENT GOALS

The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, provides a shared blueprint for peace and prosperity for people and the planet, now and into the future. At its heart are the 17 Sustainable Development Goals (SDGs), which are an urgent call for action by all countries -

ORIGIN OF SUSTAINABLE DEVELOPMENT GOALS

developed and developing - in a global partnership. They recognize that ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality, and spur economic growth – all while tackling climate change and working to preserve our oceans and forests.

THE SDGS BUILD ON DECADES OF WORK BY COUNTRIES AND THE UN, INCLUDING THE UN DEPARTMENT OF ECONOMIC AND SOCIAL AFFAIRS

- In June 1992, at the Earth Summit in Rio de Janeiro, Brazil, more than 178 countries adopted Agenda 21, a comprehensive plan of action to build a global partnership for sustainable development to improve human lives and protect the environment.

- Member States unanimously adopted the Millennium Declaration at the Millennium Summit in September 2000 at UN Headquarters in New York. The Summit led to the elaboration of eight Millennium Development Goals (MDGs) to reduce extreme poverty by 2015.

THE SDGS BUILD ON DECADES OF WORK BY COUNTRIES AND THE UN, INCLUDING THE UN DEPARTMENT OF ECONOMIC AND SOCIAL AFFAIRS

- The Johannesburg Declaration on Sustainable Development and the Plan of Implementation, adopted at the World Summit on Sustainable Development in South Africa in 2002, reaffirmed the global community's commitments to poverty eradication and the environment, and built on Agenda 21 and the Millennium Declaration by including more emphasis on multilateral partnerships.

THE UNITED NATIONS CONFERENCE ON SUSTAINABLE DEVELOPMENT

- At the United Nations Conference on Sustainable Development (Rio+20) in Rio de Janeiro, Brazil, in June 2012, Member States adopted the outcome document "The Future We Want" in which they decided, inter alia, to launch a process to develop a set of SDGs to build upon the MDGs and to establish the UN High-level Political Forum on Sustainable Development.
- The Rio +20 outcome also contained other measures for implementing sustainable development, including mandates for future programme/s of work in development financing, small island developing states and more.
- In 2013, the General Assembly set up a 30-member Open Working Group to develop a proposal on the SDGs.

THE BRUNDTLAND REPORT

- **Brundtland Report**, also called ***Our Common Future***, publication released in 1987 by the World Commission on Environment and Development (WCED) that introduced the concept of sustainable development and described how it could be achieved. Sponsored by the United Nations (UN) and chaired by Norwegian Prime Minister Gro Harlem Brundtland, the WCED explored the causes of environmental degradation, attempted to understand the interconnections between social equity, economic growth, and environmental problems, and developed policy solutions that integrated all three areas
- THE BRUNDTLAND REPORT : Sustainable development was the solution to the problems of environmental degradation discussed by the Brundtland Commission in the report Our Common Future. The remit of the Brundtland Report was to investigate the numerous concerns that had been raised in previous decades, namely, that human activity was having severe and negative impacts on the planet, and that patterns of growth and development would be unsustainable if they continued unchecked

EARTH RUMMIT AND AGENDA 21

- The concept of sustainable development formed the basis of the United Nations Conference on Environment and Development held in Rio de Janeiro in 1992.
- In June 1992, at the Earth Summit in Rio de Janeiro, Brazil, more than 178 countries adopted Agenda 21, a comprehensive plan of action to build a global partnership for sustainable development to improve human lives and protect the environment.
- The summit marked the first international attempt to draw up action plans and strategies for moving towards a more sustainable pattern of development.

THE MILLENIUM SUMMIT

- At the Millennium Declaration at the Millennium Summit in September 2000 at UN Headquarters in New York, member states unanimously adopted the MDGs.
- The Summit led to the elaboration of eight Millennium Development Goals (MDGs) to reduce extreme poverty by 2015.
- The MDGs established measurable, universally-agreed objectives for tackling extreme poverty and hunger, preventing deadly diseases, and expanding primary education to all children, among other development priorities.
- The SDGs then replaced the MDGs.

2015 - A PIVOTAL YEAR

- In January 2015, the General Assembly began the negotiation process on the post-2015 development agenda. The process culminated in the subsequent adoption of the 2030 Agenda for Sustainable Development, with 17 SDGs at its core, at the UN Sustainable Development Summit in September 2015.
- 2015 was a landmark year for multilateralism and international policy shaping, with the adoption of several major agreements:
  - Sendai Framework for Disaster Risk Reduction (March 2015)
  - Addis Ababa Action Agenda on Financing for Development (July 2015)
  - Transforming our world: the 2030 Agenda for Sustainable Development with its 17 SDGs was adopted at the UN Sustainable Development Summit in New York in September 2015.
  - Paris Agreement on Climate Change (December 2015)

SITUATION TODAY

Today, the Division for Sustainable Development Goals (DSDG) in the United Nations Department of Economic and Social Affairs (UNDESA) provides substantive support and capacity-building for the SDGs and their related thematic issues, including water, energy, climate, oceans, urbanization, transport, science and technology, the Global Sustainable Development Report (GSDR), partnerships and Small Island Developing States.

DSDG plays a key role in the evaluation of UN system wide implementation of the 2030 Agenda and on advocacy and outreach activities relating to the SDGs.

In order to make the 2030 Agenda a reality, broad ownership of the SDGs must translate into a strong commitment by all stakeholders to implement the global goals. DSDG aims to help facilitate this engagement.

CONCEPTS COVERED

- A historical Introduction to UNESCO 17 Sustainable Development Goals
- What are sustainable development goals
- Origin of sustainable sdevleopment goals
- The united nations conference on sustainable development

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Lecture 02: SDGs

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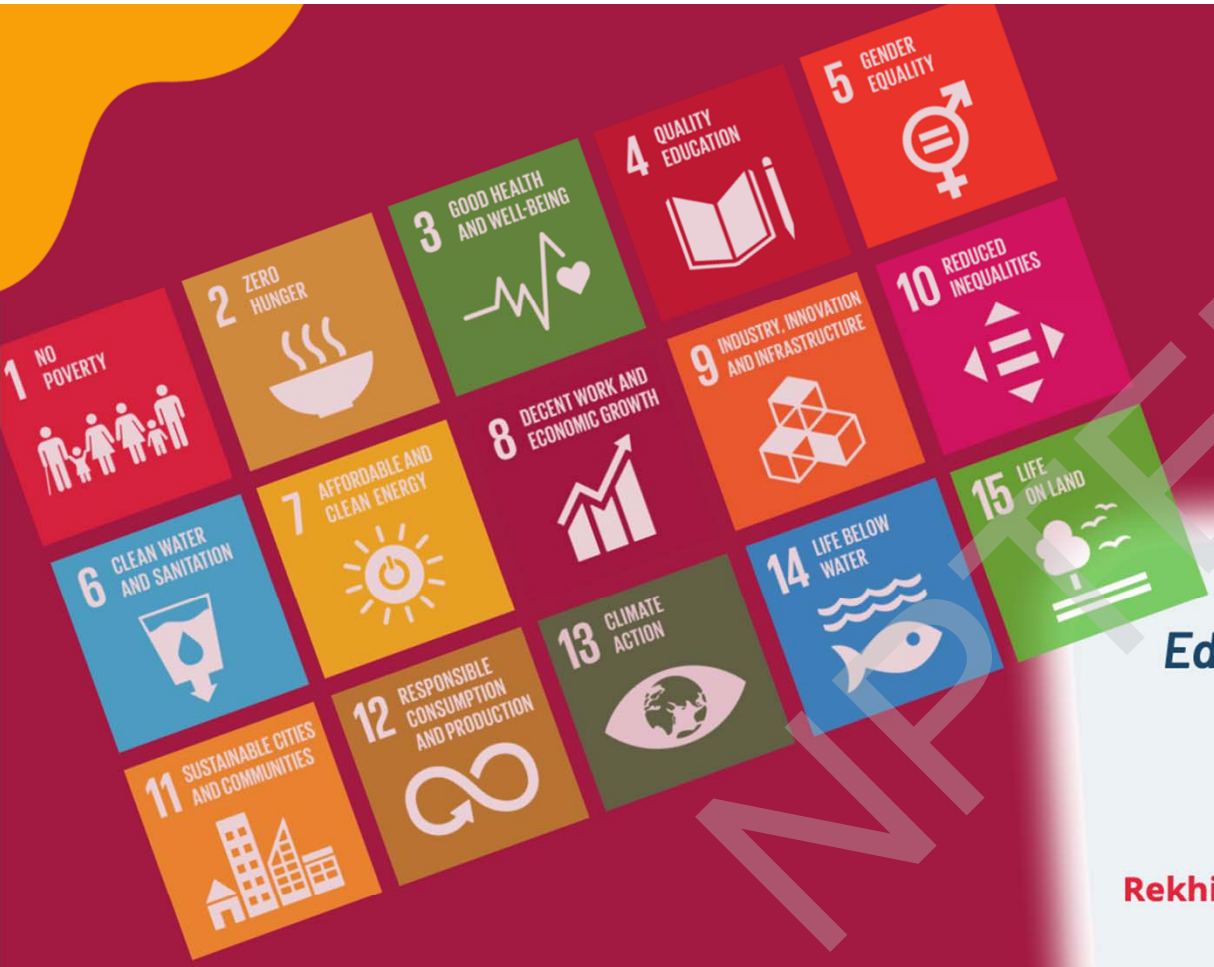

LECTURE #02

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KEYWORDS

Education

Sustainability

Sustainable Development

Multidimensional poverty

UN 17 Sustainable Development Goals

Education for Sustainable Development

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THE 17 SDG GOALS

END POVERTY IN ALL IT'S FORMS EVERYWHERE

Definition :

- Poverty can be defined as a lack of money and/or vital resources which renders it impossible or challenging for human beings to live with dignity and provide for themselves.
- Eradicating poverty in all its forms remains one of the greatest challenges facing humanity
- While the number of people living in extreme poverty dropped by more than half between1990 and 2015, too many are still struggling for the most basic human needs.

TARGETS OF GOAL

- Eradicate extreme poverty;
- reduce poverty by at least 50%
- implement social protection systems
- equal rights to ownership, basic services, technology and economic resources
- build resilience to environmental, economic and social disaster
- mobilize resources to implement policies to end poverty
- create pro-poor and gender-sensitive policy framework.

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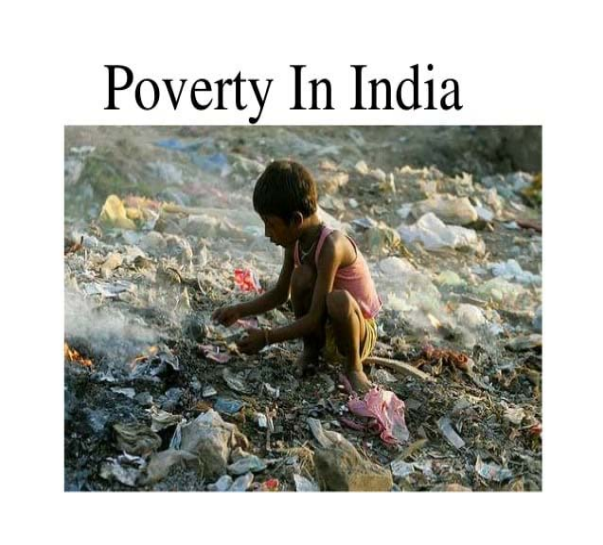
PROBLEM OF POVERTY IN INDIA

Two-thirds of people in India live in poverty: 68.8% of the Indian population lives on less than $2 a day. Over 30% even have less than $1.25 per day available - they are considered extremely poor.

This makes the Indian subcontinent one of the poorest countries in the world; women and children, the weakest members of Indian society, suffer most.

India is the second most populous country after China with about 1.2 billion people and is the seventh largest country in the world with an area of 3,287,000 km².

PROBLEM OF POVERTY IN INDIA



The highly contrasted country has enjoyed growth rates of up to 10% over many years and is one of the largest economies in the world, with a gross domestic product (GDP) of 1,644 billion US dollars.

But only a small percentage of the Indian population has benefited from this impressive economic boom so far, as the majority of people in India are still living in abject poverty.

INDIA AND GOAL 1

Global reduction in extreme poverty was driven mainly by Asia – notably China and India.

Between 2005-06 and 2015-16, the incidence of multidimensional poverty in India was almost halved, climbing down to 27.5 percent from 54.7 percent as per the 2018 global Multidimensional Poverty Index report. Within ten years, the number of poor people in India fell by more than 271 million (from 635 million to 364 million).

Traditionally disadvantaged subgroups such as rural dwellers, scheduled castes and tribes, Muslims, and young children are still the poorest in 2015-16.

However, the biggest reductions in multidimensional poverty has been witnessed among the poorest and traditionally disadvantaged groups – across states, castes, religions and age-groups.

INDIA AND GOAL 1

Multidimensional poverty among children under 10 has fallen the fastest. In 2005-06 there were 292 million poor children in India, so the latest figures represent a 47 percent decrease or a 136 million fewer children growing up in multidimensional poverty.

Multidimensional poverty measures reveal who is poor and how they are poor – the range of different disadvantages they experience; poverty level in different areas of a country and among different sub-groups of people.

The Government of India has many progressive schemes, including the world's largest employment guarantee scheme, the Mahatma Gandhi National Rural Employment Guarantee Scheme, and the National Social Assistance Programme.

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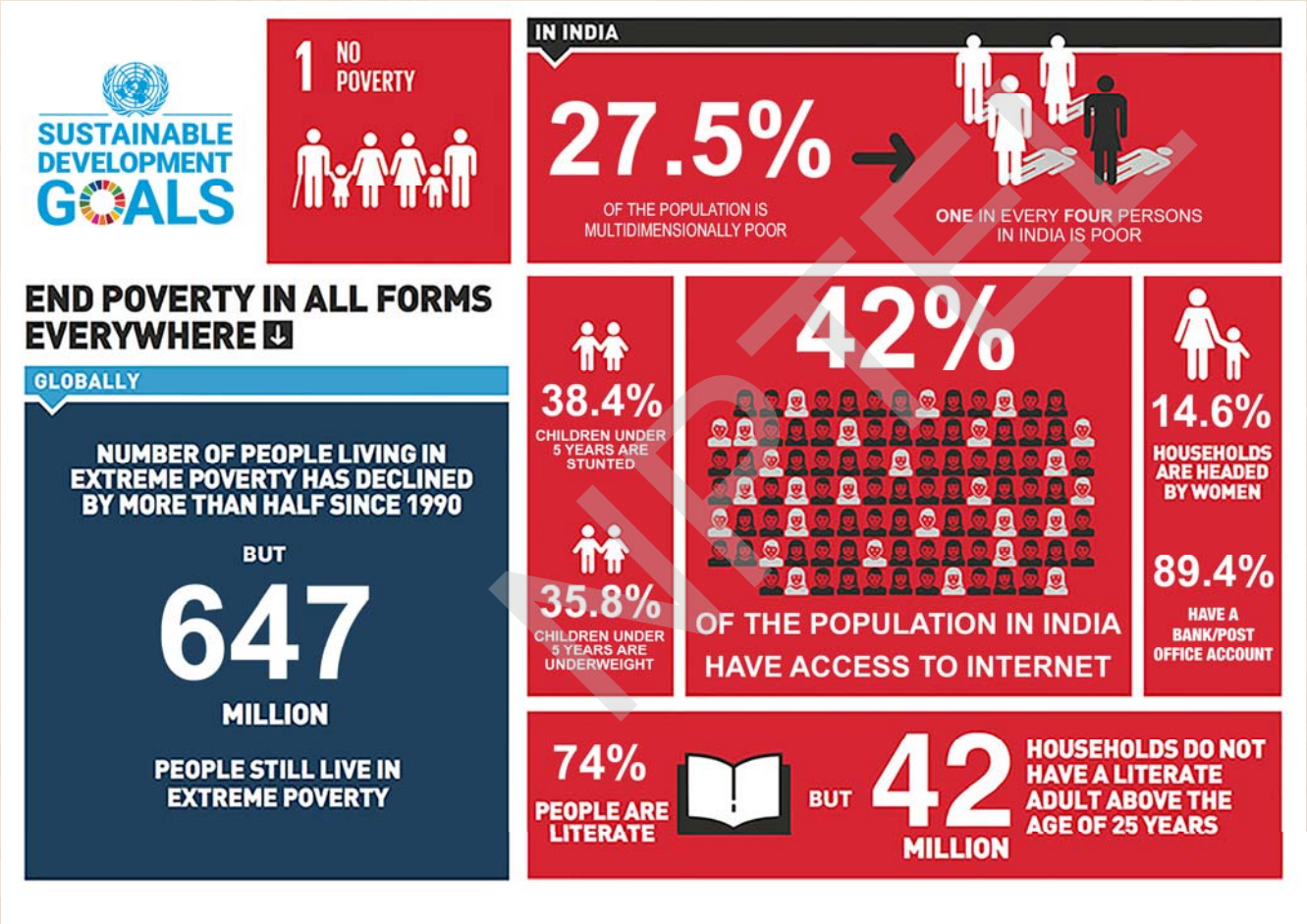

END HUNGER, ACHIEVE FOOD SECURITY & IMPROVED NUTRITION & PROMOTE SUSTAINABLE AGRICULTURE

- Malnutrition occurs when the body adapts to the prolonged absence of food, losing weight and functioning more slowly. Hunger is the leading cause of death in the world.
- Our planet has provided us with tremendous resources, but unequal access and insufficient handling leaves millions of people malnourished.
- The number of undernourished people has dropped by almost half in the past two decades because of rapid economic growth and increased agricultural productivity.

END HUNGER, ACHIEVE FOOD SECURITY & IMPROVED NUTRITION & PROMOTE SUSTAINABLE AGRICULTURE

- Many developing countries that used to suffer from famine and hunger can now meet their nutritional needs.
- The SDGs aim to end all forms of hunger and malnutrition by 2030, making sure all people, especially children, have sufficient and nutritious food all year.
- This involves promoting sustainable agriculture, supporting small-scale farmers and equal access to land, technology and markets. It also requires international cooperation to ensure investment in infrastructure and technology to improve agricultural productivity.

PROBLEM OF HUNGER IN INDIA

According to FAO estimates in 'The State of Food Security and Nutrition in the World, 2020 report, 189.2 million people are undernourished in India. By this measure 14% of the population is undernourished in India. Also, 51.4% of women in reproductive age between 15 to 49 years are anemic.

According to the report 34.7% of the children aged under five in India are stunted (too short for their age), while 20% suffer from wasting, meaning their weight is too low for their height. Malnourished children have a higher risk of death from common childhood illnesses such as diarrhea, pneumonia, and malaria.

The Global Hunger Index 2020 ranks India at 94 out of 107 countries on the basis of three leading indicators -- prevalence of wasting and stunting in children under 5 years, under 5 child mortality rate, and the proportion of undernourished in the population.

ENSURE HEALTHY LIVES & PROMOTE WELL-BEING FOR ALL AT ALL AGES

- Well-being is a feeling of satisfaction with life, a state characterized by health, happiness, and prosperity.
- Good health concerns the care of the human body and everything that can be done to protect it from sickness and intoxication and enable access to care.
- We have made great progress against several leading causes of death and disease. Life expectancy has increased dramatically, infant and maternal mortality rates have declined, and the tide against HIV and malaria deaths have halved.

ENSURE HEALTHY LIVES AND PROMOTE WELL-BEING FOR ALL AT ALL AGES

- But the world is off-track to achieve the health-related SDGs. Progress has been uneven, both between and within countries.
- While some countries have made impressive gains, national averages hide that many are being left behind.
- Multi-sector, rights-based and gender-sensitive approaches are essential to address inequalities and to build good health for all.

HEALTH & WELL BEING IN INDIA

India has made some progress in reducing its under-five mortality rate, which declined from 125 per 1,000 live births in 1990-91 to 50 per 1,000 live births in 2015-16, and its maternal mortality rate, which declined from 212 per 100,000 live births in 2007-09 to 167 in 2013.

India has also made significant strides in reducing the prevalence of HIV and AIDS across different types of high-risk categories, with adult prevalence reducing from 0.45% in 2002 to 0.27% in 2011.

However, a quarter of global TB cases occur in India where nearly 2.1 million people live with the disease, and an estimated 423,000 die annually as a result.

The Indian government's National Health Mission prioritizes national wellbeing and is leading change in this area, in addition to targeted national programs against HIV/AIDS and sexually transmitted diseases

ENSURE INCLUSIVE & EQUITABLE QUALITY EDUCATION & PROMOTE LIFELONG LEARNING OPPORTUNITIES FOR ALL

Education is the process of facilitating learning or the acquisition of knowledge, skills, values, beliefs and habits.

Quality education specifically entails issues such as appropriate skills development, gender parity, provision of relevant school infrastructure, equipment, educational materials and resources, scholarships or teaching force.

Education liberates the intellect, unlocks the imagination and is fundamental for self-respect. It is the key to prosperity and opens a world of opportunities, making it possible for each of us to contribute to a progressive, healthy society. Learning benefits every human being and should be available to all.

ENSURE INCLUSIVE & EQUITABLE QUALITY EDUCATION & PROMOTE LIFELONG LEARNING OPPORTUNITIES FOR ALL

Since 2000 there has been enormous progress in achieving the target of universal primary education. The total enrolment rate in developing regions reached 91% in 2015, and the worldwide number of children out of school has dropped by almost half. There has also been a dramatic increase in literacy rates, and many more girls are in school that ever before.

Achieving inclusive and quality education for all reaffirms the belief that education is one of the most powerful and proven vehicles for sustainable development.

It also aims to provide equal access to affordable vocational training, to eliminate gender and wealth disparities, and achieve universal access to a quality higher education

EDUCATION IN INDIA

India has made great strides in improving access to quality education, increasing elementary school enrollment and reducing the number of out-of-school children.

These achievements have been bolstered by key laws, policies and programs such as the **Right of Children to Free and Compulsory Education (RTE) Act (2009), the National Early Childhood Care and Education (ECCE) Policy (2013).** However, challenges do remain.

An estimated 6.1 million children out of school in 2014 reduced from 13.46 million in 2006. Out of 100 students, 29 per cent of girls and boys drop out of school before completing the full cycle of elementary education, and often they are the most marginalized children. (**Source: SRI-IMRB Surveys, 2009 and 2014)**

EDUCATION IN INDIA

Around 50 per cent of adolescents do not complete secondary education, while approximately 20 million children not attending pre-school. (**Source Rapid Survey of Children 2013-2014 MWCD**)

Half of primary school-going children – which constitutes nearly 50 million children – not achieving grade appropriate learning levels. (**Source: National Achievement Survey, NCERT 2017)**

Children's school readiness at age 5 has been far below expected levels. Over the first half of the Country Program, there have been various changes in the operating environment that have impacted upon how the **UNICEF India Education Program** shall operate moving forward.

ACHIEVE GENDER EQUALITY & EMPOWER ALL WOMEN AND GIRLS

Gender Equality is a principle that states that all men and women need to be treated equally and to have the same rights despite their biological differences. These should not prevent them from having the same opportunities to succeed in school and in life.

Gender bias is undermining our social fabric and devalues all of us. It is not just a human rights issue; it is a tremendous waste of the world's human potential.

ACHIEVE GENDER EQUALITY & EMPOWER ALL WOMEN AND GIRLS

By denying women equal rights, we deny half the population a chance to live life at its fullest. Political, economic and social equality for women will benefit all the world's citizens.

But there are still large inequalities in some regions, with women systematically denied the same work rights as a man.

Sexual violence and exploitation, the unequal division of unpaid care and domestic work, and discrimination in public office all remain huge barriers.

GENDER EQUALITY IN INDIA

Discrimination against women and girls is a pervasive and long-running phenomenon that characterizes Indian society at every level.

India's progress towards gender equality, measured by its position on rankings such as the Gender Development Index has been disappointing, despite fairly rapid rates of economic growth.

In the past decade, while Indian GDP has grown by around 6%, there has been a large decline in female labour force participation from 34% to 27%. The male-female wage gap has been stagnant at 50% (a recent survey finds a 27% gender pay gap in white-collar jobs).

GENDER EQUALITY IN INDIA

Crimes against women show an upward trend, in particular brutal crimes such as rapes, dowry deaths, and honor killings.

These trends are disturbing, as a natural prediction would be that with growth comes education and prosperity, and a possible decline in adherence to traditional institutions and socially prescribed gender roles that hold women back.

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bing.com/videos

- A historical Introduction to UNESCO 17 Sustainable Development Goals
- What are sustainable development goals
- Origin of sustainable development goals
- The united nations conference on sustainable development policies

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Lecture 03: SDGs(6 to 11)

KEYWORDS

- Water Crisis
- Affordable Energy
- Decent Work
- Sustainable Economic Growth
- Inequality
- Sustainable Cities

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Course objectives

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ENSURE AVAILIBILITY AND SUSTAINABLE MANAGEMENT OF WATER & SANITATION FOR ALL

- **Water is essential to life. It constitutes up to 65% of the human body and 90% of the structure of plants.**
- **Nevertheless, this fundamental element to existence is seriously threatened because of pollution of rivers, suffers shortages in certain regions of the world and costly management in cities.**
- **One in three people live without sanitation. This is causing unnecessary disease and death. Although huge strides have been made with access to clean drinking water, lack of sanitation is undermining these advances.**

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ENSURE AVAILIBILITY AND SUSTAINABLE MANAGEMENT OF WATER & SANITATION FOR ALL

- **If we provide affordable equipment and education in hygiene practices, we can stop this senseless suffering and loss of life.**
- **Safe and affordable drinking water for all by 2030 requires we invest in adequate infrastructure, provide sanitation facilities, and encourage hygiene. Protecting and restoring water-related ecosystems is essential.**

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INDIA'S WATER CRISIS - AN ALARMING URGENCY

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WATER AVAILABILITY IN INDIA

Water availability per person is dependent on population of the country and for India, per capita water availability in the country is reducing due to increase in population.

The average annual per capita water availability in the years 2001 and 2011 was assessed as 1816 cubic meters and 1545 cubic meters respectively which may further reduce to 1486 cubic meters and 1367 cubic meters in the years 2021 and 2031 respectively.

As per Ministry of Housing and Urban Affairs, 135 litre per capita per day (lpcd) has been suggested as the benchmark for urban water supply. For rural areas, a minimum service delivery of 55 lpcd has been fixed under Jal Jeevan Mission, which may be enhanced to higher level by states.

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WATER AVAILABILITY IN INDIA CONTD…..

As mentioned in the report of National Commission for Integrated Water Resources Development (NCIWRD), the percentage of water used for irrigation out of the total water use for the year 1997-98 was 83.30%.

As per NCIWRD report, the percentage of water used for irrigation out of the total water use for the year 2025 under high demand scenario was estimated as 72.48%.

This information was given by the Union Minister of State for Jal Shakti & Social Justice and Empowerment, Shri Rattan Lal Kataria in a written reply in Rajya Sabha.

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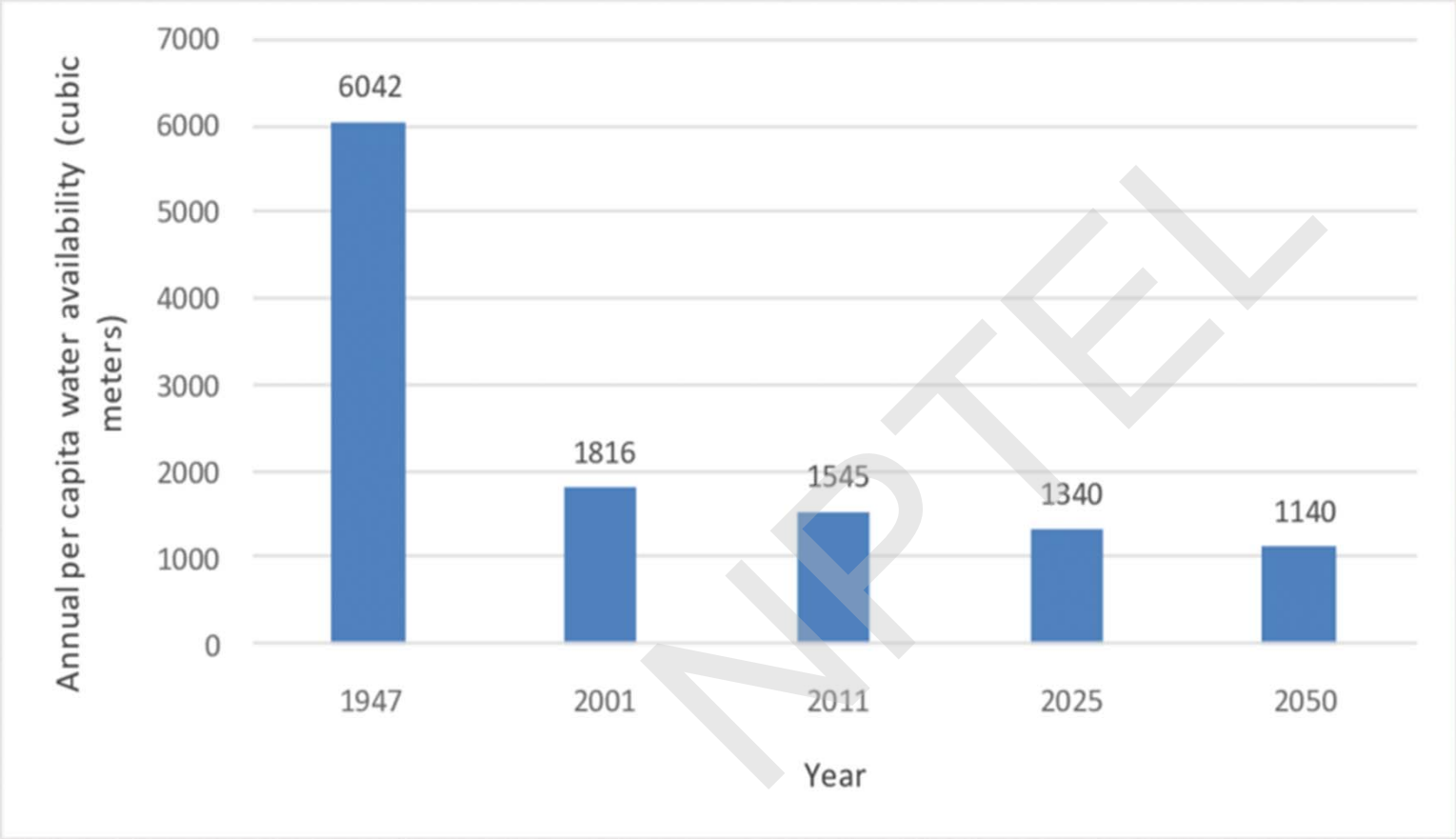
PER CAPITA WATER AVAILIBILITY IN INDIA



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ENSURE ACCESS TO AFFORDABLE, RELIABLE, SUSTAINABLE & MODERN ENERGY FOR ALL

Caption

- **Between 2000 and 2016, the number of people with electricity increased from 78 to 87 percent, and the numbers without electricity dipped to just below one billion.**

- **Yet, as the population continues to grow, so will the demand for cheap energy, and an economy reliant on fossil fuels is creating drastic changes to our climate.**

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ENSURE ACCESS TO AFFORDABLE, RELIABLE, SUSTAINABLE & MODERN ENERGY FOR ALL

- **Investing in solar, wind and thermal power, improving energy productivity, and ensuring energy for all is vital if we are to achieve SDG 7 by 2030.**

- **Expanding infrastructure and upgrading technology to provide clean and more efficient energy in all countries will encourage growth and help the environment.**

Caption

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ENERGY AVAILABILITY IN INDIA

India has seen extraordinary successes in its recent energy development, but many challenges remain, and the Covid-19 pandemic has been a major disruption.

In recent years, India has brought electricity connections to hundreds of millions of its citizens; promoted the adoption of highly-efficient LED lighting by most households; and prompted a massive expansion in renewable sources of energy, led by solar power.

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ENERGY AVAILABILITY IN INDIA CONTD….

The gains for Indian citizens and their quality of life have been tangible. However, the Covid-19 crisis has complicated efforts to resolve other pressing problems.

These include a lack of reliable electricity supply for many consumers; a continued reliance on solid biomass, mainly firewood, as a cooking fuel for some 660 million people; financially ailing electricity distribution companies, and air quality that has made Indian cities among the most polluted in the world

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INDIA'S SOLAR POWER REVOLUTION

Solar power is set for explosive growth in India, matching coal's share in the Indian power generation mix within two decades in the STEPS – or even sooner in the Sustainable Development Scenario. As things stand, solar accounts for less than 4% of India's electricity generation, and coal close to 70%.

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INDIA'S SOLAR POWER REVOLUTION

By 2040, they converge in the low 30%s in the STEPS, and this switch is even more rapid in other scenarios. This dramatic turnaround is driven by India's policy ambitions, notably the target to reach 450 GW of renewable capacity by 2030, and the extraordinary cost-competitiveness of solar, which out-competes *existing* coal-fired power by 2030 even when paired with battery storage.

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INDIA'S SOLAR POWER REVOLUTION

The rise of utility-scale renewable projects is underpinned by some innovative regulatory approaches that encourage pairing solar with other generation technologies, and with storage, to offer "round the clock" supply. Keeping up momentum behind investments in renewables also means tackling risks relating to delayed payments to generators, land acquisition, and regulatory and contract uncertainty.

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INDIA'S SOLAR POWER REVOLUTION

However, the projections in the STEPS do not come close to exhausting the scope for solar to meet India's energy needs, especially for other applications such as rooftop solar, solar thermal heating, and water pumps.

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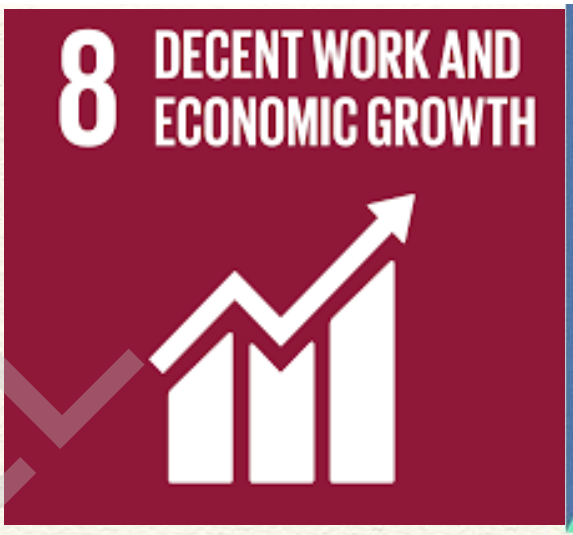
PROMOTE SUSTAINED INCLUSIVE & SUSTAINABLE ECONOMIC GROWTH, FULL & PRODUCTIVE EMPLOYMENT & DECENT WORK FOR ALL



Over the past 25 years the number of workers living in extreme poverty has declined dramatically, despite the lasting impact of the 2008 economic crisis and global recession. In developing countries, the middle class now makes up more than 34 percent of total employment – a number that has almost tripled between 1991 and 2015.

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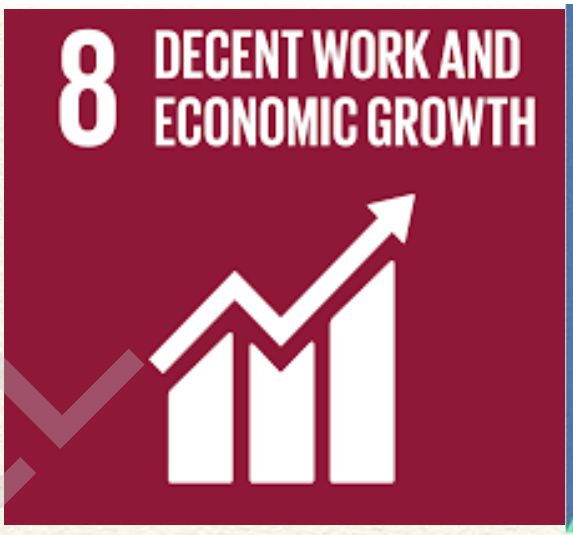
PROMOTE SUSTAINED INCLUSIVE & SUSTAINABLE ECONOMIC GROWTH, FULL & PRODUCTIVE EMPLOYMENT & DECENT WORK FOR ALL



However, as the global economy continues to recover we are seeing slower growth, widening inequalities, and not enough jobs to keep up with a growing labour force. According to the International Labour Organization, more than 204 million people were unemployed in 2015.

The SDGs promote sustained economic growth, higher levels of productivity and technological innovation. Encouraging entrepreneurship and job creation are key to this, as are effective measures to eradicate forced labour, slavery and human trafficking. With these targets in mind, the goal is to achieve full and productive employment, and decent work, for all women and men by 2030.

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WHAT IS SUSTAINABLE ECONOMIC GROWTH ?

Economic growth occurs when real output increases over time. Real output is measured by Gross Domestic Product (GDP) at constant prices, so that the effect of price rises on the *value* of national output is removed.

Sustainable economic growth means a rate of growth which can be maintained without creating other significant economic problems, especially for future generations.

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WHAT IS SUSTAINABLE ECONOMIC GROWTH ?

There is clearly a *trade-off* between rapid economic growth today, and growth in the future. Rapid growth today may exhaust resources and create environmental problems for future generations, including the depletion of oil and fish stocks, and global warming.

Periods of growth are often triggered by increases in aggregate demand, such as a rise in consumer spending, but sustained growth must involve an increase in output. If output does not increase, any extra demand will push up the price level.

NPTEL

BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE & SUSTAINABLE INDUSTRIALIZATION, AND FOSTER INNOVATION

Caption

- **Investment in infrastructure and innovation are crucial drivers of economic growth and development. With over half the world population now living in cities, mass transport and renewable energy are becoming ever more important, as are the growth of new industries and information and communication technologies.**

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BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE & SUSTAINABLE INDUSTRIALIZATION, AND FOSTER INNOVATION

Caption

- Technological progress is also key to finding lasting solutions to both economic and environmental challenges, such as providing new jobs and promoting energy efficiency. Promoting sustainable industries, and investing in scientific research and innovation, are all important ways to facilitate sustainable development.
- More than 4 billion people still do not have access to the Internet, and 90 percent are from the developing world. Bridging this digital divide is crucial to ensure equal access to information and knowledge, as well as foster innovation and entrepreneurship.

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INDIA AND GOAL 9

The government's flagship interventions like MAKE IN INDIA and START UP INDIA as well as Pandit Deendayal Upadhyay Shramev Jayate Karyakram are fueling innovation and sustainable industrial and economic development.

Targets

Develop quality, reliable, sustainable and resilient infrastructure, including regional and trans-border infrastructure, to support economic development and human well being, with a focus on affordable and equitable access for all.

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INDIA AND GOAL 9

Promote inclusive and sustainable industrialization and, by 2030, significantly raise industry's share of employment and gross domestic product, in line with national circumstances, and double its share in least developed countries.

Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets.

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INDIA AND GOAL 9

By 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes, with all countries taking action in accordance with their respective capabilities.

Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries, in particular developing countries, including, by 2030, encouraging innovation and substantially increasing the number of research and development workers per 1 million people and public and private research and development spending.

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INDIA AND GOAL 9

Facilitate sustainable and resilient infrastructure development in developing countries through enhanced financial, technological and technical support to African countries, least developed countries, land-locked developing countries and small island developing states.

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SUSTAINABLE DEVELOPMENT GOALS

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE AND SUSTAINABLE INDUSTRIALISATION, AND FOSTER INNOVATION

GLOBALLY

14.2%

OF THE WORLD'S WORKFORCE EMPLOYED IN MANUFACTURING

1.1 MANUFACTURING JOB CREATES 2.2 JOBS IN OTHER SECTORS

IN INDIA

111 MILLION

PEOPLE EMPLOYED IN MICRO, SMALL AND MEDIUM ENTERPRISES PRODUCE

33%

OF THE MANUFACTURING OUTPUT

GDP GROWTH AVERAGED

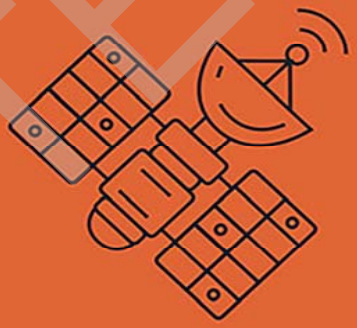
7.2%

BETWEEN 2018 - 2019

566 MILLION INTERNET SUBSCRIBERS (MOBILE AND LANDLINE)

CO_2

1.80 MT ONE OF THE LOWEST PER CAPITA CO2 EMISSIONS IN THE WORLD



MANGALYAAN IS THE WORLD'S LEAST EXPENSIVE INTERPLANETARY MISSION TO MARS

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REDUCE INEQUALITY WITHIN AND AMONG COUNTRIES

Caption

INCOME INEQUALITY

- **Income inequality is on the rise—the richest 10 percent have up to 40 percent of global income whereas the poorest 10 percent earn only between 2 to 7 percent. If we take into account population growth inequality in developing countries, inequality has increased by 11 percent.**
- **Income inequality requires global solutions. This involves improving the regulation and monitoring of financial markets and institutions, encouraging development assistance and foreign direct investment to regions where the need is greatest. Facilitating the safe migration and mobility of people is also key to bridging the widening divide.**

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REDUCE INEQUALITY WITHIN AND AMONG COUNTRIES

Caption

OTHER FORMS OF INEQUALITY

Inequalities are not only driven and measured by income, but are determined by other factors - gender, age, origin, ethnicity, disability, sexual orientation, class, and religion. These factors determine inequalities of opportunity which continue to persist, within and between countries. In some parts of the world, these divides are becoming more pronounced. Meanwhile, gaps in newer areas, such as access to online and mobile technologies, are emerging. The result is a complex mix of internal and external challenges that will continue to grow over the next twenty-five years.

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PROBLEM OF INEQUALITY IN INDIA

The problem with the growth-centred narrative is that while growth is necessary for poverty alleviation, it is not sufficient.

India has an inequality problem. In 2020–21, India's economy experienced the largest annual contraction (7.3%; NSO, MOSPI 2021) in the post-Independence period due to the health and economic fallouts of the Covid-19 crisis.

Yet, the stock market has recovered from an initial dip and is up by 75% from a year ago. The total number of Indian billionaires rose to 140 last year from 102 and their combined wealth nearly doubled to $596 billion. India has the third largest number of billionaires in the world after the US and China. Yet, the number of poor in India is estimated to have increased by 75 million due to Covid-19, accounting for nearly 60% of the global increase in poverty.

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PROBLEM OF INEQUALITY IN INDIA

The problem with the growth-centred narrative is that while growth is necessary for poverty alleviation, it is not sufficient.

After accounting for population and economic differences across countries, an earlier analysis showed that India does have more than its expected share of multimillionaires (Ghatak and Ray, 2014). Additionally, India's share of the world's extreme poor is higher than its share of the world population. India accounts for 139 million of the total 689 million people (20.17%) living in extreme poverty in 2017 (World Bank 2020), with its population being 17.8% of the world population (World Bank 2019).

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MAKE CITIES AND HUMAN SETTLEMENT INCLUSIVE, SAFE, RESILIENT & SUSTAINABLE

Caption

- More than half of us live in cities. By 2050, two-thirds of all humanity—6.5 billion people—will be urban. Sustainable development cannot be achieved without significantly transforming the way we build and manage our urban spaces.
- The rapid growth of cities—a result of rising populations and increasing migration—has led to a boom in mega-cities, especially in the developing world, and slums are becoming a more significant feature of urban life.

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MAKE CITIES AND HUMAN SETTLEMENT INCLUSIVE, SAFE, RESILIENT & SUSTAINABLE

Caption

- **Making cities sustainable means creating career and business opportunities, safe and affordable housing, and building resilient societies and economies. It involves investment in public transport, creating green public spaces, and improving urban planning and management in participatory and inclusive ways.**

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INDIA AND GOAL 11

Targets

By 2030, ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums.

By 2030, provide access to safe, affordable, accessible and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention given to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons.

By 2030, enhance inclusive and sustainable urbanization and capacity for participatory, integrated and sustainable human settlement planning and management in all countries. Strengthen efforts to protect and safeguard the world's cultural and natural heritage.

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INDIA AND GOAL 11

Targets

By 2030, significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global gross domestic product caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations.

By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management.

By 2030, provide universal access to safe, inclusive and accessible, green and public spaces, in particular for women and children, older persons and persons with disabilities.

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https://www.theindiaforum.in/article/does-india-have-inequality-problem

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CONCEPTS COVERED

- SDG 6
- SDG 7
- SDG 8
- SDG 9
- SDG 10
- SDG 11
- Application of SDG's in the Indian context
- Understanding the problem of water scarcity in India

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NPTEL ONLINE CERTIFICATION COURSES

Education for Sustainable Development

Prof. Atasi Mohanty

Rekhi Centre of Excellence for the Science of Happiness,

IIT Kharagpur

Lecture 04: SDGs

Course objectives

- Students will gain understanding of Sustainable Development and its 17 goals
- Students will be able to understand and explain SDG 4- Quality Education for all
- Students will be able to analyze how to use Education for Sustainable Development to achieve SDG 4.7 and SDG 4.4
- Students will be able to study and link sustainable education and global partnership

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Expected Learning outcomes

- By the end of the course students will be able to:
- Describe the UNESCO's 17 Sustainable Development Goals
- Identify elements of Education for Sustainable Development
- Illustrate and integrate ESD to achieve SDG 4.7
- Explain and formulate ESd to accomplish SDG 4.4
- Interpolate sustainable education and global partnership

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ENSURE SUSTAINABLE CONSUMPTIONS & PRODUCTION PATTERNS

- Achieving economic growth and sustainable development requires that we urgently reduce our ecological footprint by changing the way we produce and consume goods and resources. Agriculture is the biggest user of water worldwide, and irrigation now claims close to 70 percent of all freshwater for human use.
- The efficient management of our shared natural resources, and the way we dispose of toxic waste and pollutants, are important targets to achieve this goal. Encouraging industries, businesses and consumers to recycle and reduce waste is equally important, as is supporting developing countries to move towards more sustainable patterns of consumption by 2030.

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ENSURE SUSTAINABLE CONSUMPTIONS AND PRODUCTION PATTERNS

- A large share of the world population is still consuming far too little to meet even their basic needs. Halving the per capita of global food waste at the retailer and consumer levels is also important for creating more efficient production and supply chains. This can help with food security, and shift us towards a more resource efficient economy.

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What does sustainable consumption mean?

Ideally, sustainable consumption means that our food practices meet the dietary needs of today in terms of nutrition and satiation, without compromising the needs of future generations (especially regarding ecological concerns). Additionally, a crucial part of sustainability is that sustainable consumption must enable socio-economic justice for everyone involved in the supply chain, as much as it addresses consumer and environmental concerns.

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TAKE URGENT ACTION TO COMBAT CLIMATE CHANGE AND ITS IMPACTS

- There is no country that is not experiencing the drastic effects of climate change. Greenhouse gas emissions are more than 50 percent higher than in 1990. Global warming is causing long-lasting changes to our climate system, which threatens irreversible consequences if we do not act.
- The annual average economic losses from climate-related disasters are in the hundreds of billions of dollars. This is not to mention the human impact of geo-physical disasters, which are 91 percent climate-related, and which between 1998 and 2017 killed 1.3 million people, and left 4.4 billion injured. The goal aims to mobilize US$100 billion annually by 2020 to address the needs of developing countries to both adapt to climate change and invest in low-carbon development.

Caption

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TAKE URGENT ACTION TO COMBAT CLIMATE CHANGE AND ITS IMPACTS

- Supporting vulnerable regions will directly contribute not only to Goal 13 but also to the other SDGs. These actions must also go hand in hand with efforts to integrate disaster risk measures, sustainable natural resource management, and human security into national development strategies. It is still possible, with strong political will, increased investment, and using existing technology, to limit the increase in global mean temperature to two degrees Celsius above pre-industrial levels, aiming at 1.5°C, but this requires urgent and ambitious collective action.

Caption

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Impact of climate change on the Indian subcontinent

Since the middle of the twentieth century, India has witnessed a rise in average temperature; a decrease in monsoon precipitation; a rise in extreme temperature and rainfall events, droughts, and sea levels; and an increase in the intensity of severe cyclones, alongside other changes in the monsoon system. There is compelling scientific evidence that human activities have influenced these changes in regional climate.

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Impact of climate change on the Indian subcontinent

Human-induced climate change is expected to continue apace during the twenty-first century. To improve the accuracy of future climate projections, particularly in the context of regional forecasts, it is essential to develop strategic approaches for improving the knowledge of Earth system processes, and to continue enhancing observation systems and climate models.

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CONSERVE AND SUSTAINABLY USE OCEANS, SEAS AND MARINE RESOURCES FOR SUSTAINABLE DEVELOPMENT

Caption

- The world's oceans – their temperature, chemistry, currents and life – drive global systems that make the Earth habitable for humankind. How we manage this vital resource is essential for humanity as a whole, and to counterbalance the effects of climate change.
- Over three billion people depend on marine and coastal biodiversity for their livelihoods. However, today we are seeing 30 percent of the world's fish stocks overexploited, reaching below the level at which they can produce sustainable yields.

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CONSERVE AND SUSTAINABLY USE OCEANS, SEAS AND MARINE RESOURCES FOR SUSTAINABLE DEVELOPMENT

Caption

- Oceans also absorb about 30 percent of the carbon dioxide produced by humans, and we are seeing a 26 percent rise in ocean acidification since the beginning of the industrial revolution. Marine pollution, an overwhelming majority of which comes from land-based sources, is reaching alarming levels, with an average of 13,000 pieces of plastic litter to be found on every square kilometer of ocean.
- The SDGs aim to sustainably manage and protect marine and coastal ecosystems from pollution, as well as address the impacts of ocean acidification. Enhancing conservation and the sustainable use of ocean-based resources through international law will also help mitigate some of the challenges facing our oceans.

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PROTECT RESTORE AND PROMOTE SUSTAINABLE USE OF TERRITORIAL ECOSYSTEMS, SUSTAINABLE MANAGE FORESTS, COMBAT CERTIFICATION AND HALT AND REVERSE LAND DEGRADATION AND HALT BIODIVERSITY LOSS

Caption

- Human life depends on the earth as much as the ocean for our sustenance and livelihoods. Plant life provides 80 percent of the human diet, and we rely on agriculture as an important economic resources. Forests cover 30 percent of the Earth's surface, provide vital habitats for millions of species, and important sources for clean air and water, as well as being crucial for combating climate change.
- Every year, 13 million hectares of forests are lost, while the persistent degradation of dry lands has led to the desertification of 3.6 billion hectares, disproportionately affecting poor communities.

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PROTECT RESTORE AND PROMOTE SUSTAINABLE USE OF TERRITORIAL ECOSYSTEMS, SUSTAINABLE MANAGE FORESTS, COMBAT CERTIFICATION AND HALT AND REVERSE LAND DEGRADATION AND HALT BIODIVERSITY LOSS

Caption

- While 15 percent of land is protected, biodiversity is still at risk. Nearly 7,000 species of animals and plants have been illegally traded. Wildlife trafficking not only erodes biodiversity, but creates insecurity, fuels conflict, and feeds corruption.

- Urgent action must be taken to reduce the loss of natural habitats and biodiversity which are part of our common heritage and support global food and water security, climate change mitigation and adaptation, and peace and security.

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PROMOTE PEACEFUL AND INCLUSIVE SOCIETIES FOR SUSTAINABLE DEVELOPMENT, PROVIDING ACCESS TO JUSTICE FOR ALL AND BUILD EFFECTIVE, ACCOUNTABLE AND INCLUSIVE INSTITUTIONS AT ALL LEVELS

Caption

- We cannot hope for sustainable development without peace, stability, human rights and effective governance, based on the rule of law. Yet our world is increasingly divided. Some regions enjoy peace, security and prosperity, while others fall into seemingly endless cycles of conflict and violence. This is not inevitable and must be addressed.

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PROMOTE PEACEFUL AND INCLUSIVE SOCIETIES FOR SUSTAINABLE DEVELOPMENT, PROVIDING ACCESS TO JUSTICE FOR ALL AND BUILD EFFECTIVE, ACCOUNTABLE AND INCLUSIVE INSTITUTIONS AT ALL LEVELS

Caption

- Armed violence and insecurity have a destructive impact on a country's development, affecting economic growth, and often resulting in grievances that last for generations. Sexual violence, crime, exploitation and torture are also prevalent where there is conflict, or no rule of law, and countries must take measures to protect those who are most at risk

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PROMOTE PEACEFUL AND INCLUSIVE SOCIETIES FOR SUSTAINABLE DEVELOPMENT, PROVIDING ACCESS TO JUSTICE FOR ALL AND BUILD EFFECTIVE, ACCOUNTABLE AND INCLUSIVE INSTITUTIONS AT ALL LEVELS

Caption

- The SDGs aim to significantly reduce all forms of violence, and work with governments and communities to end conflict and insecurity. Promoting the rule of law and human rights are key to this process, as is reducing the flow of illicit arms and strengthening the participation of developing countries in the institutions of global governance.

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STRENGTHEN THE MEANS OF IMPLEMENTATION AND REVITALISE THE GLOBAL PARTNERSHIP FOR SUSTAINABLE DEVELOPMENT

- The SDGs can only be realized with strong global partnerships and cooperation. Official Development Assistance remained steady but below target, at US$147 billion in 2017. While humanitarian crises brought on by conflict or natural disasters continue to demand more financial resources and aid. Many countries also require Official Development Assistance to encourage growth and trade.

Caption

NPTEL

STRENGTHEN THE MEANS OF IMPLEMENTATION AND REVITALIZE THE GLOBAL PARTNERSHIP FOR SUSTAINABLE DEVELOPMENT

Caption

- The world is more interconnected than ever. Improving access to technology and knowledge is an important way to share ideas and foster innovation. Coordinating policies to help developing countries manage their debt, as well as promoting investment for the least developed, is vital for sustainable growth and development.
- The goals aim to enhance North-South and South-South cooperation by supporting national plans to achieve all the targets. Promoting international trade, and helping developing countries increase their exports is all part of achieving a universal rules-based and equitable trading system that is fair and open and benefits all.

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https://in.one.un.org/page/sustainable-development-goals/sdg-9/

https://www.theindiaforum.in/article/does-india-have-inequality-problem

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CONCEPTS COVERED

- A historical Introduction to UNESCO 17 Sustainable Development Goals
- What are sustainable development goals
- Origin of sustainable development goals
- The united nations conference on sustainable development

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NPTEL ONLINE CERTIFICATION COURSES

Education for Sustainable Development

Prof. Atasi Mohanty
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IIT Kharagpur

Lecture 5: SDG Case Studies

CONCEPTS COVERED

- UN SDG GOALS : A CASE BY CASE STUDY
- CASE STUDY : TIMOR LESTE
- CASE STUDY : BOSNIA AND HERZGOVNIA
- CASE STUDY : ALBANIA

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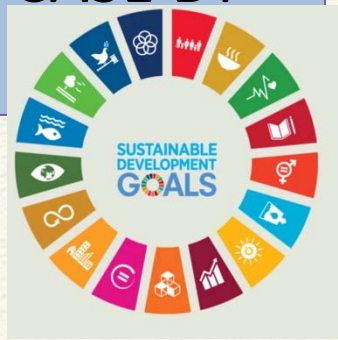
UN SDG GOALS - A GLOBAL COMPARATIVE CASE BY CASE STUDY



Case studies are an in-depth investigation of one particular individual, group, time-period or event. They encompass a range of qualitative and quantitative research tools to investigate underlying principles of an occurrence within a real-life context.

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UN SDG GOALS - A GLOBAL COMPARATIVE CASE BY CASE STUDY

Benefits

- **Comprehensive**. The most significant benefit of case studies is that they enable a holistic review. Unlike standalone research techniques which give more of a snapshot, e.g. surveys, a case study offers the opportunity for a researcher to use a range of tools on one subject. This gives time and space to build a detailed understanding of the topic, establishing a sound platform from which to explore the factors influencing the case study in greater detail.

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UN SDG GOALS - A GLOBAL COMPARATIVE CASE BY CASE STUDY

- **Reducing bias**. Case studies capture a range of perspectives, as opposed to the single view of an individual you get with a survey response or interview. This gives the opportunity to gain a greater understanding of the subject in hand and reduces the potential for any bias, by diluting the agenda of a particular individual.

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UN SDG GOALS - A GLOBAL COMPARATIVE CASE BY CASE STUDY

The methodology for the district heating case studies included face-to-face interviews with up to five members of staff from relevant social housing departments and external organizations where third party management organizations were used. The methodology also featured in-depth technical and financial analysis and a focus group with residents from each of the case study schemes.

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Case study of TIMOR-LESTE

SDGs ADDRESSED-

SDG 2 - ZERO HUNGER

SDG 3 - GOOD HEALTH AND WELL BEING

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Case study of TIMOR-LESTE

The programme on “promoting sustainable food and nutrition security in Timor-Leste” (the Programme) was undertaken in late 2009 to address the conditions contributing to chronic and acute undernutrition through a harmonized approach utilizing both technical support and long-term capacity building. Through establishing a public-private partnership, micro-nutrient rich food could be locally produced, enhancing the fragile private sector whilst simultaneously addressing chronic maternal and child malnutrition.

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Case study of TIMOR-LESTE

THE SITUATION-

- As one of the world's youngest countries, Timor-Leste experienced various hindrances to societal development. The underdevelopment of public and private sectors such as education, health and economy created a cycle of disadvantage and thus a multi-sectorial approach was vital to any effective development initiative. For example, the high prevalence of illiteracy, with 49.1 per cent of women and 38.7 per cent of men illiterate in Tetum among persons over the age of 15, hindered economic participation for both genders, with women also facing additional challenges due to cultural norms. A lack of economic mobility manifested in restricted land ownership rights, capital and credit access.

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Case study of TIMOR-LESTE

THE SITUATION-

- This environment also led to the emergence of an informal private sector, where a series of micro-enterprises had developed, primarily dominated by women and their families selling small products like household goods. This alternate sector was not participating in the growth of the national economy as the activities were often unpaid or paid in non-monetary forms. There were various structural factors which damaged private sector development in Timor-Leste, including the absence of an effective legal framework and a reluctance to invest in the seemingly fragile young republic.

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Case study of TIMOR-LESTE

THE SITUATION-

- A major developmental issue affecting Timor-Leste was stunting occurring due to chronic malnutrition, especially that of children under five years and young mothers. The rates of stunting were at a startling high of 58 per cent when the Programme began.

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THE STRATEGY

With the objective of addressing child and maternal Moderate Acute Malnutrition (MAM), part of the Programme focused on the local production of a supplementary feeding product. Through the public-private partnership with food production company, Timor Global, the Timor-Leste Ministry of Health and WFP, the Programme was able to provide a sustainable solution to acute malnutrition whilst building local capacity and improving food storage issues.

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THE STRATEGY

The food produced, Timor Vita, was a substance specifically tailored to the taste and needs of the people of Timor-Leste. It was tested by local people and packaged in an easily transportable manner.

The three-way partnership was an example of a social business established in Timor-Leste. Each partner had an integral role to ensure it benefited those targeted and allowed the growth of local production, in order to diminish reliance on foreign imports and produce a better tasting and fresher product.

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THE STRATEGY

WFP initially provided technical assistance through managing the conversion of the food production plant and training factory staff and local farmers to ensure quality standards were met. The Ministry of Health purchased the processing and packing equipment necessary to set up the factory. Timor Global was the local production partner. The UN partners also contributed a generator to stabilize production and the Government of Timor- Leste invested US$ 2 million in the project in 2012.

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THE STRATEGY

Timor Vita not only supplied women and children with an accessible, fresh supplementary food but also encouraged local economic growth through the promotion of local agriculture and the creation of employment opportunities for around 40 workers within the food production factory. Although the costs of purchasing Timor Vita were on par with those of imported substances, it had a longer shelf life and therefore saved money by preventing wastage.

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Part of the Programme focused on the local production of a supplementary feeding product

Timor Vita also encouraged local economic growth

The three-way partnership was an example of a social business

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ACHIEVEMENTS:

- With support from the programme, the Ministry of Health revised the 2004 National Nutrition Strategy in order to strengthen the food security and nutrition sectors.
- The prevalence of underweight under-five children dropped from 45.7% in 2009 to 28.9% in 2012, according to data from the Health Management Information System.

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ACHIEVEMENTS:

- A total of 84 school gardens were created in four programme districts, with the schools using the harvested vegetables to supplement school meals. Some 450 members of school communities where gardens were planted in 2011 received nutrition training the following year.

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ACHIEVEMENTS:

- The production of Timor Vita, a micronutrient powder (MNP) supplementation, was launched in April 2012, and more than 1,200 children aged 6-23 months received the supplementation.

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Energy efficiency and renewable energy in Bosnia and Herzegovina

SDG Addressed -

SDG GOAL 7 - Affordable and clean energy

PROBLEM:

Bosnia and Herzegovina had made significant strides in economic stabilization and national cohesion in the two decades since the break-up of the former Yugoslavia. However, the environment

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Energy efficiency and renewable energy in Bosnia and Herzegovina

sector is critically stagnant: there is a lack of environmental policy and legislation, poorly developed management and implementation capacities, little public participation in environmental decision-making and a lack of reliable information and data.

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Energy efficiency and renewable energy in Bosnia and Herzegovina

STRATEGY

- Providing capacity for developing Local Environmental Action Plans;
- Providing seed funding for local service delivery priorities;
- Raising awareness and national level support for environmental action through an environmental innovation fund; and
- Creating systems for capturing environmental data.

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Energy efficiency and renewable energy in Bosnia and Herzegovina

In order to facilitate such matters, comprehensive planning mechanisms were designed to provide innovative actions based on strategic-planning documents. Besides emphasising the ability of stakeholders from local to national levels to play a positive role, they were founded on inclusive environmental governance principles and the idea that successful environmental services require both horizontal and vertical cooperation.

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ACHIEVEMENTS:

- With the programme's support, 37 new municipal Action Plans were drawn up to underpin the development of local environmental governance capacity and introduce planning methodologies. Altogether 12,418 people (LEAP Coordinators, Local Action Groups, consultants, NGOs, public/private companies, citizens, etc.) participated in a variety of LEAP activities through public meetings and questionnaire surveys.

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ACHIEVEMENTS:

- Municipalities demonstrated their effective resource management as partners in the micro-capital grants scheme, distributing 19 grants (up to 50% of the project cost) in support of actions identified in the Plans to solve the most pressing problems.

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ACHIEVEMENTS:

- Local level developments, lessons and best practice were used to influence policy development and mobilize awareness on environmental issues. Nationally, the complex legal and institutional background for environmental governance was reviewed and a ‘road map’ prepared. A Designated National Authority for the Kyoto Protocol was established, and a gap analysis for an Environmental Information System was prepared for BiH’s future environmental administration.

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BETTER WATER AND SANITATION SERVICES THROUGH CONSUMER RIGHTS BASED CONTRACT IN ALBANIA

SDG'S ADDRESSED:

SDG GOAL 5 : GENDER EQUALITY

SDG GOAL 8 : DECENT WORK AND ECONOMIC GROWTH

SDG GOAL 16 : PEACE, JUSICE AND STRONG INSTITUTIONS

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BETTER WATER AND SANITATION SERVICES THROUGH CONSUMER RIGHTS BASED CONTRACT IN ALBANIA

THE PROBLEM:

The sustainable supply of safe, sufficient and quality water us a priority for the government of Albania. While access to water has improved considerably in recent years, poor service is a persistent issue: the water pipes are old, the quality of water is low and there are frequent water cuts. The decentralization and commercialization of water supply services have pushed water prices up, but improvements in service are lagging behind.

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BETTER WATER AND SANITATION SERVICES THROUGH CONSUMER RIGHTS BASED CONTRACT IN ALBANIA

the joint UNDP and World Bank programme, "Economic governance, regulatory reform, public participation, and pro- poor development: 2009 – 2012" (the Programme):

was designed to build the capacities of Government, utility regulators and consumer associations to effectively manage, regulate and monitor key public utility sectors such as electricity and water, and

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BETTER WATER AND SANITATION SERVICES THROUGH CONSUMER RIGHTS BASED CONTRACT IN ALBANIA

to strengthen the voice of consumers to promote and protect their rights and encourage accessible and affordable services. As such, it assisted Albania in its efforts to reduce regional disparities, promote social inclusion, meet its Millennium Development Goals targets and prepare for accession to the European Union.

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ACHIEVEMENTS:

- The programme assisted in the development of a model water contract to regulate the relationship between consumers and Albania's 56 water utility companies and eventually improve the quality of water and sewerage services. This model contract will impact the lives of over three million people.
- A consumer complaints management system was established to receive and manage consumer complaints. A public web site for the Consumer Protection Commission was also developed.

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ACHIEVEMENTS:

- The Joint Programme supported the drafting of a methodology for prioritizing rural investments as well as the development of business plans for the electricity supplier (KESH) and the transmitter (OST).
- It also established mechanisms for participatory debate and decision-making as a way to improve access to electricity and water services, and thus to reduce poverty.

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REFERENCES

- https://www.sdgfund.org/case-studies

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